

WORKSHOP

Asset Management

On a merchant BESS fleet, the gap between theoretical revenue — perfect-foresight dispatch — and what you actually realize is where asset management lives.

60–90 min, modular. Each block = ~10 min story + ~10 min open discussion.

Interactive; no China-specific focus; no dashboards without real data and credible sources; practical over conceptual; everything adaptable overseas.

Anonymization rules, applied throughout: the operating fleet is "a fleet of four grid-scale BESS assets in a liberalized spot market"; no province/market names for that fleet; counterparties unnamed; ratios and percentages preferred over absolute currency. GB/international market data (Modo Energy, Exelon — named, credible) provides the international grounding.

Cold open

"The gap between theoretical and realized revenue on a BESS fleet is where asset management actually lives. We measure it per asset, per day. Today I'll walk through where it leaks, in four places, with real numbers — and each time I'd rather hear your version than finish mine."

Frame slide: the four layers — Physical Asset / Data & Settlement / Market & Dispatch / Commercial & Lifecycle. "Value is won or lost in all four. The hardware is the smallest part of the variance."

~20 min

Your P&L is wrong until proven otherwise

L2 · DATA &
SETTLEMENT

- ~80% of asset-management work is data plumbing. Nobody believes this until they own a bill.

- Real case: a mis-signed charge component — a sign-convention mismatch between counterparty formats — persisted for months: 79 charge records rebuilt, plus duplicate discharge records removed. Found only by re-deriving every settlement from first principles. End state: eight months of books reconciled to the cent.
- Settlement formats change without notice; parsers drift silently. Manual recon $\Rightarrow$ month-long detection latency; automated per-bill re-derivation $\Rightarrow$ a day.

Exhibit: one anonymized bill re-derivation walkthrough (printed charge $\rightarrow$ stored sign $\rightarrow$ reconciled monthly total). No market names.

DISCUSSION

"Who in your organization re-derives the settlement statement before it gets paid? And what is your detection latency when a counterparty changes format?"

Expected pushback: "Our market's settlement is simpler." $\rightarrow$ "Then your version of this failure lives in the metering and telemetry layer — same question: who re-derives it?"

~20 min

L3 ·
MARKET &
DISPATCH

The capture gap is the job

- Metric first: capture rate = realized $\div$ theoretical (perfect-foresight LP) revenue, per asset per day. Everything in dispatch quality lives in that number.
- Real case: a legacy forecast model used day-ahead prices published *after* real-time trading closes — apparent capture inflated ~100%. Exposed only by running multiple models side-by-side. Lesson: model governance — never one unchallenged model.
- Second real case: AI intelligence feeds have real unit costs — one carries a weekly credit quota (~36–40 calls measured vs a 140-call wish). Question cadence is now rotated to budget, with runs aborted on the first quota error. AI operations are an economics problem, not a demo.

Exhibit: capture-by-model comparison table (theoretical vs realized, per model — one visibly inflated). Percentages only.

DISCUSSION

"What do you measure dispatch quality against — and has anyone audited the benchmark itself?"

~20 min

L4 ·
COMMERCIAL
& LIFECYCLE

Attribution is the trust layer

- "The market was bad" is not an explanation; it's a mood. Per-asset daily P&L attribution decomposes performance into price / dispatch / degradation drivers — the only thing that makes a performance-linked contract verifiable.
- Real practice: daily per-asset attribution; strategy-comparison workflow (actual dispatch vs alternatives, P&L waterfall); portfolio roll-up; IRR re-run on operating data — acquisition models go stale fast.
- Commercial consequence: performance contracts (capture floor + upside share) sell only on top of attribution both parties trust. Contract design and measurement design are the same document.

Exhibit: one anonymized daily P&L waterfall for a single asset (price / dispatch / other, decomposed). Indexed units, no currency labels.

DISCUSSION

"What would your owners accept as proof of outperformance — and does your current reporting survive that standard?"

~20 min

CROSS-
CUTTING

Small team, big stack — the AI-leverage operating model

- Staffing reality: a small senior team covers all four layers because four domain agents do the analyst legwork — market strategist, economics quant, trading-ops analyst, investment aggregator.
- What makes agents trustworthy in production: strict grounding (answers only from data returned by tools — never the model's general knowledge), persistent domain memory across sessions, human review loops.
- And the honest limits: quotas, grounding failures — and the day a vendor silently changed their login flow, and a nightly intelligence feed started spamming magic-link emails instead of data. Resilience engineering is part of AI operations.

Exhibit: architecture sketch — four agents over one data platform, with the grounding/memory loop. Optional live read-only walk if the room allows (real data, no credentials).

DISCUSSION

"Where would an agent break first in your data environment — and who catches it?"

10 min

The take-home checklist

CLOSE

- Do you re-derive your settlement, or accept it? (L2)
- Do you measure dispatch against a counterfactual — and has the counterfactual been audited? (L3)
- Does your P&L decompose into drivers your owners would accept? (L4)
- What in your stack is instrumented, attributed, *proven* — and what is still asserted?

"The hardware is the smallest part of the variance. The stack is the job."

Speaker prep notes

- **Timing flex:** 60-min version — drop the Block 4 discussion (keep the story) and compress the close.
- **Evidence pack to refresh before presenting:**
 - anonymized bill re-derivation exhibit (from the settlement audit trail);
 - capture-by-model table (last quarter, percentages);
 - daily P&L attribution waterfall for one asset (indexed);
 - agent architecture sketch (4 agents, grounding rules, memory loop);
 - quota/economics numbers for the AI-ops anecdote.
- **Sources named in the room:** Modo Energy + Elexon for GB market context (credible, public/subscription); fleet data stays anonymized.

Do NOT: name the fleet's market or provinces, name counterparties, or show any credential-adjacent screen. If asked "which market is the fleet in?" — "a liberalized spot market; the failure modes are the point, and they're market-agnostic."

Real operating cases from a four-asset grid-scale BESS fleet and a 29-market daily data pipeline; generalized for an international room. Framework companion: *The BESS Asset Value Stack*.